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LATE PALEOZOIC MAGNETIZATION OF MINERALIZED AND UNMINERALIZED ORDOVICIAN CARBONATES FROM EAST TENNESSEE: EVIDENCE FOR A POST-ORE CHEMICAL EVENT

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Abstract. Thermal demagnetization studies on samples from mineralized and unmineralized rocks in the Lower Ordovician Knox Group of east Tennessee reveal a mean direction of D/I = 157°/+18° (α_{95} = 7.7°, k = 52.9) after 60% unfolding, combining stable endpoint and remagnetization circle data. Although not significant at the 95% confidence level, the fold test is suggestive of a synfolding (Alleghenian) age of the magnetization. The magnetization cannot be postfolding, i.e., Permian or younger, because the in situ directions do not resemble those for post-Carboniferous times. The magnetization is unlikely to be much older than the folding because the directions with respect to bedding yield a paleopole characteristic of the Early Permian, but unlike poles for pre-Late Carboniferous times. Characteristic magnetizations from mineralized and unmineralized collapse breccias are in good agreement and fail the conglomerate test. The best estimate for the paleopole resulting from this study falls at 40°N, 126°E, within the Late Carboniferous segment of the North American Paleozoic apparent polar wander path. Thus the magnetization is interpreted as synfolding and secondary due to widespread Kiaman remagnetization. Blocking temperatures of up to 520°C and the acquisition of isothermal remanent magnetization suggest magnetite as the carrier of the magnetization. Scanning electron microscopy and energy dispersive analysis of magnetic extracts from limestone samples reveal abundant magnetite spheres. Secondary dolomites have the same magnetic characteristics as the primary limestones, and it is inferred that similar magnetite spheres are the carriers of their magnetization. Because this magnetization, and therefore its carrier, occurs in sphalerite and secondary dolomite, it must be coeval with or younger than the ore, which itself is thought to be of middle to late Paleozoic age. It is argued that the magnetization was acquired during growth of magnetite, associated with a late Paleozoic chemical event.

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Introduction

Mississippi Valley-type (MVT) deposits, which consist of various combinations of sphalerite, galena, fluorite, and barite in carbonate rocks, are generally considered to have formed from brines expelled from sedimentary basins [Anderson and MacQueen, 1982]. Along with hydrocarbon accumulations, MVT deposits provide important information on the timing of basinal fluid migration and the tectonic evolution of the source basins [e.g., Oliver, 1986]. Thus one of the most important goals of research on MVT deposits has been the determination of their age. In all but a few cases, largely in Missouri [York et al., 1981; Stein and Kish, 1985], MVT deposits lack minerals containing K, Rb, and U and cannot be dated by conventional radiometric methods. This has led in the past to attempts to date the deposits using paleomagnetic methods [e.g., Beales et al., 1974]. Subsequent efforts of this type, mostly confined to the economically important southeast Missouri districts, demonstrated the presence of a late Paleozoic magnetization in ores as well as in the host rocks [Wu and Beales, 1981; Wisniowiecki et al., 1983]. On the basis of geochemical considerations, Wisniowiecki et al. [1983] argued that the magnetization carriers predated the formation of the southeast Missouri ore deposits. Results from the present study suggest that the magnetization in the east Tennessee MVT deposits is actually post-ore in age.

We report here the results of a paleomagnetic investigation of the Lower Ordovician Knox Group and its MVT ores in the Mascot-Jefferson City, Copper Ridge, and Sweetwater districts. The east Tennessee area of MVT mineralization is the largest and best known example of the Appalachian-style MVT deposits [Hoagland, 1971, 1976; Ohle, 1980, 1985], which exhibit important differences from the previously investigated southeast Missouri MVT deposits. For instance, whereas the southeast Missouri deposits are related to an intracratonic basin and are hosted by essentially flat-lying sediments [Sverjensky, 1981], the Appalachian-type deposits are related to basins that formed on the margin of the craton and are hosted by rocks that were deformed after mineralization [Hoagland, 1976]. Furthermore, the southeast Missouri MVT deposits contain large amounts of Pb, whereas the Appalachian-type deposits are notable for their lack of Pb [Sangster, 1983].

The Appalachian-type deposits extend from Tennessee through Pennsylvania to Newfoundland [Hoagland, 1971] and therefore appear to represent a continent-wide basin dewatering

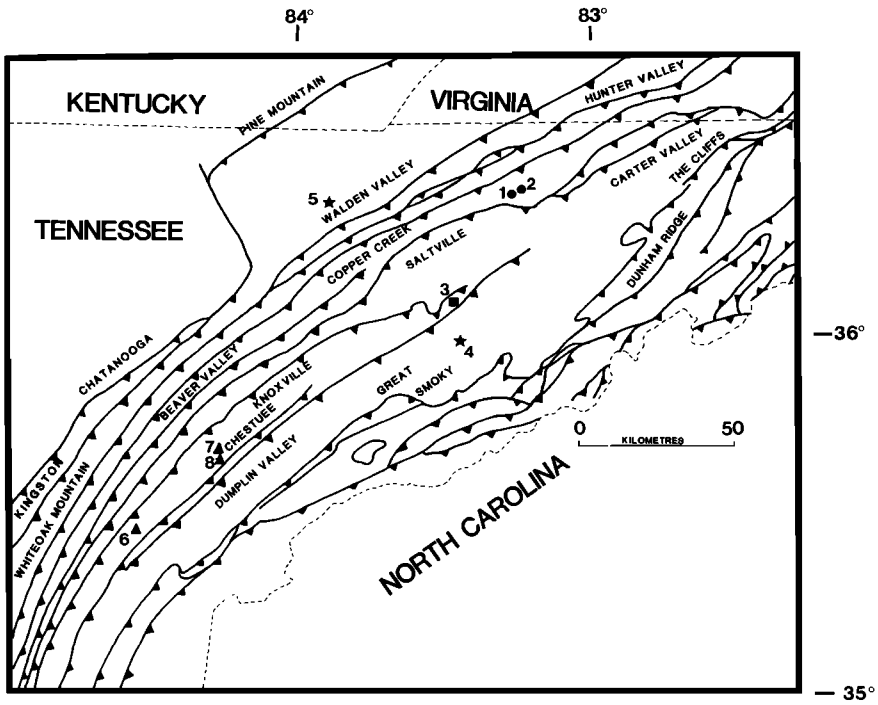


Fig. 1. Map showing the distribution of regional sampling sites (stars) and sites in the Copper Ridge (solid circles), Mascot-Jefferson City (squares), and Sweetwater (triangles) districts (1, Copper Ridge highway exposure; 2, Idol mine; 3, New Market mine; 4, Douglas Dam exposure; 5, Lost Creek prospect; 6, Stephens Pit; 7, Ballard Quarry; 8, Fork Creek pit). For a schematic stratigraphic cross section see Figure 2. Also shown are major thrust sheets.

process for which there is, as yet, no good age control. Thus we initiated a paleomagnetic study of these important mineralized regions in the hope that it could provide information on the timing of the magnetization relative to the mineralization and large-scale Appalachian tectonic processes.

Distribution and Geologic Characteristics of MVT Mineralization in East Tennessee

Appalachian-style MVT mineralization is widespread in the Lower Ordovician Knox Group of east Tennessee, although mineable deposits are confined to the Mascot-Jefferson City, Copper Ridge, and Sweetwater districts (Figure 1). Mineralization at Mascot-Jefferson City and Copper Ridge consists almost entirely of sphalerite and dolomite, whereas that at Sweetwater consists of fluorite and barite with small amounts of sphalerite and dolomite [Maher, 1971]. Fluid inclusions in these minerals indicate that mineralization took place at temperatures of about 100°-170°C at Mascot-Jefferson City [Taylor et al., 1983] and 70°-150°C at Sweetwater [Zimmerman and Kesler, 1981]. Salinities of most fluids were about 20 equivalent wt % NaCl in both districts, although they ranged to lower values at Sweetwater. The high CO_2 and Ca^{++} contents of fluid inclusions, in conjunction with carbonate equilibria [Haynes et al., 1987], indicate relatively acidic conditions, where the simultaneous transport of zinc and reduced sulfur would have been possible [Sverjensky, 1981]. Haynes et al. have argued that the fluid inclusion data are consistent with a model for ore formation involving the release of "abnormal" fluid pressures, most likely deep in the host basin rather than at the basin margin.

The Mascot-Jefferson City, Sweetwater, and Copper Ridge ores are found in solution collapse breccia bodies (Figure 2) up to 300 m below an extensive unconformity at the top of the Knox Group [Harris, 1971]. This unconformity was the locus of widespread karsting during Middle Ordovician time and considerable interest has focused on the possibility that the ore-hosting breccias were produced by this karsting [Laurence, 1971]. Steinberg and Kopp [1986] have suggested that mineralization transecting the Knox unconformity in the Lost Creek area (location 5 in Figures 1 and 2) represents a near-surface manifestation of karsting processes in the underlying Knox and that breccia systems should exist which connect the surface karst zones to the larger "hydrothermal" breccias up to 300 m below the unconformity. Karst breccias formed during this early Middle Ordovician exposure are referred to as fine rock matrix breccias and consist of fragments of primary dolomite in a matrix of unstratified dolomite sand and silt [Hoagland et al., 1965]. The Douglas Dam locality (location 4) represents karst-related breccias of this type, sampled in the uppermost dolomites of the Mascot, within 50 m of the Knox unconformity (Figure 2). The fine rock matrix breccias are distinguishable from breccias related to hydrothermal activity, many of which are cemented by secondary dolomite,

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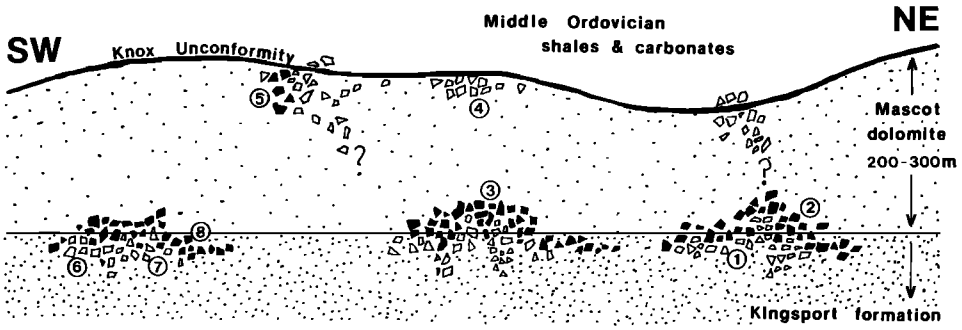


Fig. 2. Palinspastic (schematic) cross section of the Ordovician rocks below the Knox unconformity in east Tennessee, from site 6 (SW) to site 2 (NE) as located in Figure 1. Section is not to scale, and site locations have been projected onto this profile across thrust sheets. Pre-ore fine rock matrix breccias are depicted by open breccia symbols, whereas mineralized, "hydrothermal" breccias are shown by solid symbols.

fluorite, or sphalerite [Hoagland et al., 1965; McCormick et al., 1971]. The significance of these observations, particularly with respect to the possibility that the ore-matrix breccias do not owe their present characteristics and extent to karsting, remains unresolved [Haynes, 1986]. Thus one of our goals in this paleomagnetic survey was to test both types of breccia and to evaluate the age of magnetization relative to brecciation.

The presence of main-stage sphalerite as detrital material deposited and lithified within the breccia bodies prior to fracturing and folding [Kendall, 1960; Matlock and Misra, 1986] appears to provide indisputable evidence that the sphalerite predates Alleghenian tectonism. Thus we have the following sequence of events: Early Ordovician deposition of the host carbonates of the Knox Group, Middle Ordovician karstification, and a subsequent (?) hydrothermal brecciation event followed by sphalerite-barite-fluorite mineralization which is bracketed in age between the Middle Ordovician and the Late Carboniferous to Early Permian (Alleghenian) thrusting and folding. This paper will address the following questions: (1) When in this sequence did the magnetic oxides form? and (2) When was the magnetization acquired?

Sampling and Laboratory Techniques

Oriented hand samples (at least four per site) were taken in the New Market mine (Jefferson City-Mascot district) and from surface outcrops in the Copper Ridge and Sweetwater districts (Figure 1). Surface samples were collected to estimate the effects of weathering on the magnetization by comparison with unweathered samples from underground mines. A wide variety of rock types was sampled, ranging from unaltered limestones and primary dolomites to recrystallized dolostones and ore.

Sampling of breccia blocks of varying size from mineralized and unmineralized collapse structures as well as stratified sediments from subsurface and surface exposures provided the opportunity to test the relative ages of magnetization in ore and host rocks. Different bedding attitudes of the three districts allowed fold tests to establish the timing of magnetization with respect to regional deformation.

Bearing and plunge of the reference plane of the hand samples were determined with a Brunton compass and the average bedding strike and dip were recorded from each site. Three to four 2.54-cm-diameter cores were drilled in the laboratory from each hand sample and then cut to specimens of 2.1 cm length (about 10 cm^3). After preparation and during the laboratory treatment, specimens were kept inside a shielded room (maximum field approximately 200 nT=nanotesla) to reduce the effects of viscous behavior. The natural remanent magnetization of the samples was

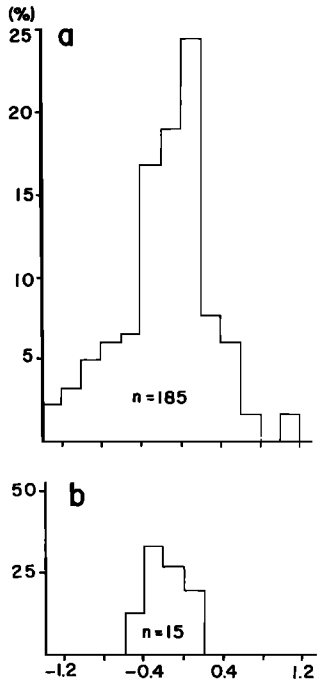


Fig. 3. (a) Frequency distribution of initial intensities of the natural remanent magnetization (NRM) of all samples. (b) Histogram of intensity distribution of mineralized, unmineralized, and gangue dolomite samples from the New Market mine only (locality 3, Figures 1 and 2).

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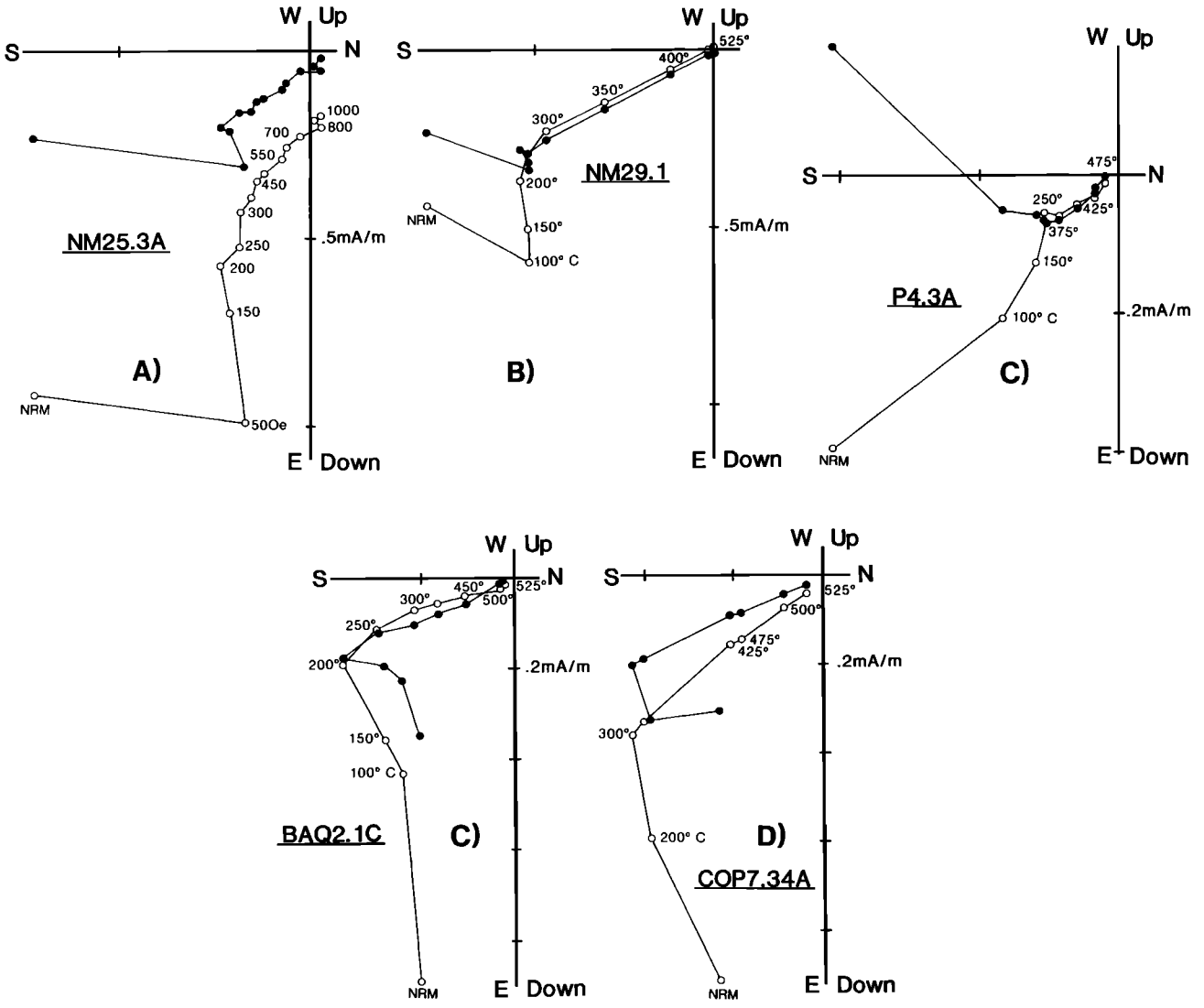


Fig. 4. (a) Characteristic orthogonal alternating field and (b-e) thermal demagnetization diagrams [Zijderveld, 1967] of samples from the New Market mine (Figures 4a, 4b, and 4c), Sweetwater (Figure 4d) and Copper Ridge districts (Figure 4e) before tilt correction. Note the close directional similarity between characteristic directions from recrystallized (Figures 4a-4c) and unaltered (Figures 4d and 4e) samples. Solid (open) circles represent projections on the horizontal (vertical) plane.

measured using a ScT cryogenic magnetometer. Stepwise progressive alternating field (af) demagnetization experiments were performed using a GSD-1 Schonstedt demagnetizer and progressive thermal demagnetization was conducted in a Schonstedt furnace with a residual field of less than 5 nT. Selected representative samples were subjected to detailed thermal and af demagnetization experiments to establish the optimum treatment for the bulk of the collection. Results were plotted in orthogonal vector diagrams [Zijderveld, 1967] and in stereographic projections, and were evaluated by visual inspection followed by principal component analysis [Kirschvink, 1980]. Demagnetization results were rejected for sample directions with a mean angular deviation (MAD) [Kirschvink, 1980] greater than 10°. Mean directions were then calculated by combining stable endpoint and remagnetization circle data [Bailey and Halls, 1984].

Paleomagnetic Results

The rocks investigated for this study display uniformly high intensities of natural remanent magnetization (J_{mean} = 1.29 mA/m, Figure 3), and exhibit uniform demagnetization behavior (Figure 4), regardless of lithology (limestone or dolomite), geographic distribution (subsurface or surface outcrop), or relationship to the mineralization (mineralized breccia or unmineralized host rock). Demagnetization by alternating fields (Figure 4a) was not adequate to separate or remove all individual magnetic components. Due to the overlap of the coercivity

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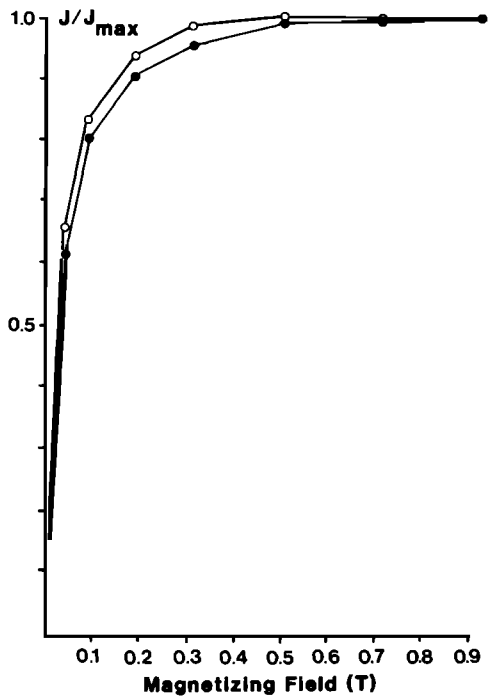


Fig. 5. Acquisition of isothermal remanent magnetization (IRM) of a surface sample (solid circles) and a subsurface sample (open circles).

spectra of magnetic components in most of the samples, af demagnetization succeeded only in a few cases in isolating a remanence with apparently univectorial characteristic directions. Better results were achieved with stepwise thermal demagnetization (Figures 4b-4e). Removal of a steep component at temperatures up to 300°C isolated a single component trending toward the origin. The low-stability component is probably a viscous remanent magnetization (VRM) acquired in the present-day geomagnetic field. Unblocking temperatures for the higher-temperature component up to 520°C and the almost complete saturation of the rocks below 0.3 T during the acquisition of isothermal remanent magnetization (IRM) (Figure 5) suggest that relatively pure magnetite is the predominant carrier of the magnetization. The sharp directional changes at 300°C and the univectorial trend of the characteristic remanent magnetization (ChRM) toward the origin (Figures 4b-4e) suggest that the magnetic components have been separated successfully.

Site mean directions based on stable endpoint analysis are listed in Table 1. The similarities in the directions determined from mineralized and unmineralized rocks and from both stratified carbonates and breccia components strongly argue for a secondary, postmineralization and postbrecciation age of the magnetization. We will return to this aspect after a discussion of a few cases displaying less typical magnetic behavior and a description of our observations of the carriers of magnetization.

Dolomites from the Douglas Dam locality (location 4 in Figures 1 and 2), located just below the Early to Mid-Ordovician unconformity [Harris, 1971], show an additional complexity (Figure 6). Two types of unmineralized dolomite were sampled at Douglas Dam: (1) unaltered karst-brecciated dolomite and (2) stratified primary dolomite. Curved trajectories on Zijderveld plots (Figures 6a-6c) indicate the presence of at least two higher-temperature components with overlapping unblocking temperature spectra, which cannot be resolved using conventional techniques. Plotting the results of progressive thermal demagnetization on stereographic projections (Figures 6d-6f) demonstrates the lack of stable endpoints and reveals that the changes of the resulting vectors follow great circle paths. IRM acquisition experiments on these particular samples (Figure 7) indicate the presence of a high-coercivity phase (likely hematite), with saturation not reached at maximum fields of 1.4 T. The application of the remagnetization circle method [Halls, 1976], however, enabled resolution of well-defined remagnetization paths. Poles to remagnetization circles are shown as squares in Figure 8a and yield a poorly defined common intersection pointing to the north and up, although the polarity of the component thus determined cannot be defined using the great circle method. Uniform bedding attitudes at Douglas Dam, moreover, preclude resolution of more than one component of magnetization in these dolomites, as discussed by Halls [1976]. Because the breccias at this locality are farther away from known MVT mineralization than the other sites studied and are believed to relate directly to the karstification event at the Early to Middle Ordovician boundary [Harris, 1971], one could argue that the high coercivity component is related to Ordovician weathering (and formation of hematite). An Ordovician age of this magnetization, however, would require a correction for the structural tilt, as measured above the unconformity. It can be seen from Table 1 that when the full tilt correction is applied, the direction becomes shallowly upward, which is unlike the expected Ordovician directions. However, the direction coincides well with the other (stable endpoint) directions observed throughout the study area. Thus we interpret these directions to be secondary as well.

The overall mean direction and associated statistical parameters (Table 1) have been calculated using the algorithm of Bailey and Halls [1984] which combines stable endpoint directions and great circle data. Figures 8a and 8b show the poles to remagnetization planes and the sample directions, respectively, before bedding correction. Site mean directions and the resulting overall mean direction are shown in situ (Figure 8c) and after bedding correction (Figure 8d).

In general, all the breccias investigated show magnetic behavior and characteristic directions which are indistinguishable from those of host rocks (see Table 1). A breccia body in the Sweetwater district (Ballard pit, locality 7) provides the lone exception. Although the magnetic behavior of the samples from this locality is indistinguishable from material sampled elsewhere, the directions of the breccia samples (not included in Table 1) are distributed

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TABLE 1. Summary of Paleomagnetic Data From the Ordovician Knox Group

		Before Unfolding		After 100% Unfolding		After 60% Unfolding			
Locality	N	D, deg	I, deg	D, deg	I, deg	D, deg	I, deg	k	α_{95}
Copper Ridge district									
1, Copper Ridge	7	150.9	51.7	149.3	15.3	150.1	30.1	54.7	8.2
2, Idol Mine	4	144.2	45.4	144.8	-1.6	145.2	17.2	30.9	16.8
New Market-Jefferson City district									
3, New Market	12	154.4	14.8	154.1	9.0	154.2	11.3	25.4	8.8
Regional sites, outside districts									
4, Douglas Dam	17	160.7	36.4	160.2	-3.6	160.6	12.4	14.5	17.2
5, Lost Creek	8	160.1	33.1	157.5	17.0	158.3	23.5	6.7	23.2
Sweetwater district									
6, Stephens Pit	4	155.3	26.1	151.8	20.5	153.1	22.8	20.5	20.8
7, Ballard Qry	3	166.0	30.7	167.0	15.8	167.1	22.0	14.4	33.8
8, Fork Creek	2	166.6	15.5	166.3	-4.4	166.3	3.6	--	--
Mean*		157.9	31.9 k=31.6 α_{95}=10.0	156.6	8.7 k=39.8 α_{95}=8.9	157.0	18.0 k=52.9 α_{95}=7.7		
Pole positions		(32.8°N, 121.7°E)		(44.1°N, 129.8°E)		39.9°N	126.3°E		

N, number of samples; D, declination; I, inclination; k, Fisher's [1953] estimate of precision; α_{95}, semiangle of the cone of 95% confidence.

* Mean of eight site means and the best estimate direction from great-circle remagnetization circles (according to the method of Bailey and Halls, 1984).

randomly, which could suggest a positive conglomerate test. This would have major implications for interpretation of the age of the brecciation relative to that of the magnetization. A check on this apparently positive conglomerate test result is possible because original bedding plane features can be observed in the individual breccia blocks. Although rotation of blocks during brecciation would randomize the declinations, a concentrated inclination distribution should be realized upon correction for the bedding tilt. Because tops and bottoms of these individual breccia blocks cannot be determined, only the absolute value of the inclinations (regardless of sign) can be used for a conglomerate test. Bedding corrections for the individual breccia components did not improve the clustering of the inclinations, therefore suggesting that the magnetization is not primary and did not predate brecciation. We conclude that the random directions in the Ballard pit, which contrast sharply with the clustered in situ directions of the other breccias studied, provide an inconclusive breccia test and that their origin must be ascribed to the presence of incoherent magnetizations or possibly to near-surface collapse during recent weathering.

Magnetic Extracts

Saturation magnetization intensity and rock magnetic behavior indicate that only small amounts (approximately 1 ppm) of magnetic materials are present in our samples. Moreover, these materials are likely to be of submicroscopic size (up to 0.5 μm for pseudo-single domain grains) which renders optical in situ identification of the magnetic carriers difficult. Therefore scanning electron microscopy (SEM) was combined with energy dispersive analysis (EDA) of magnetic extracts to identify the magnetic carriers and to estimate their chemical composition.

Of the lithologies examined in this study,

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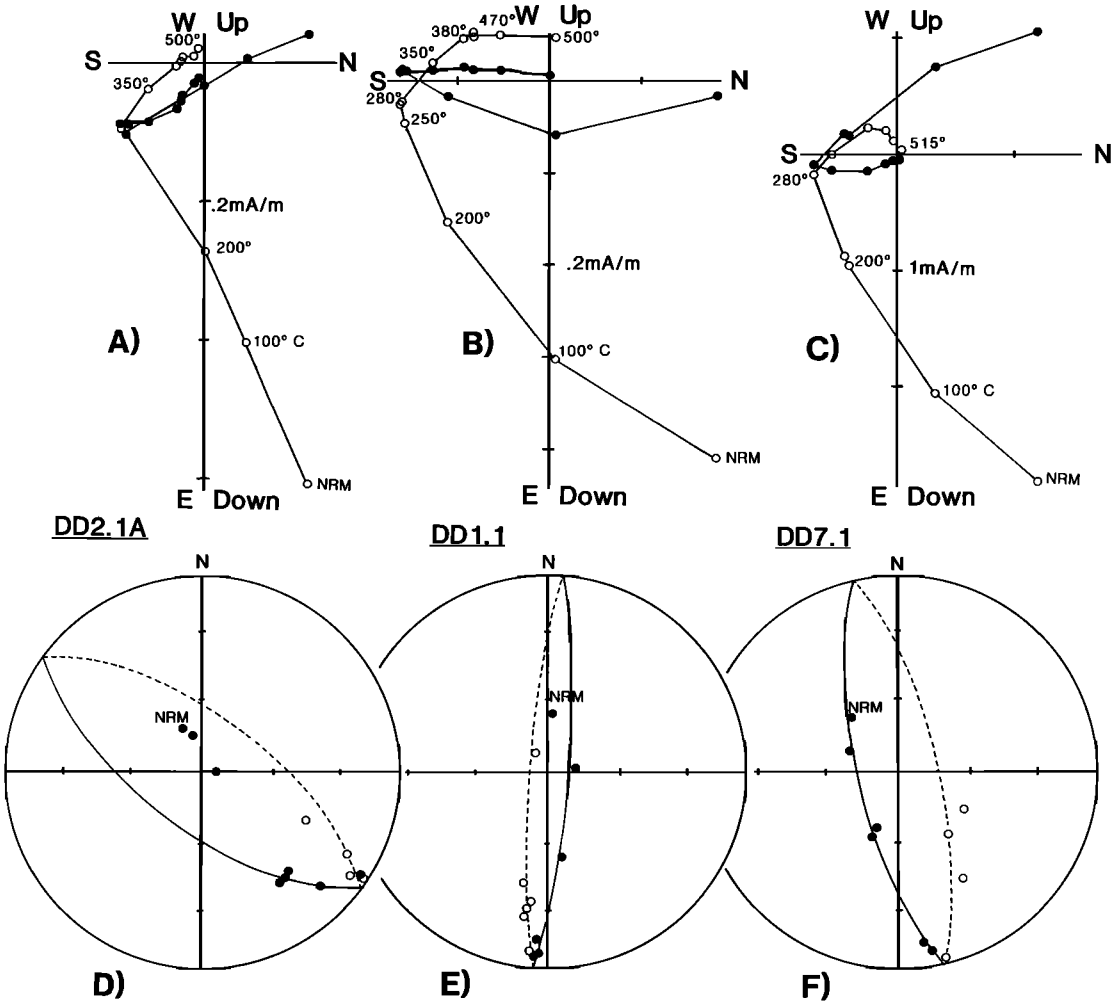


Fig. 6. (a)-(c) Typical orthogonal thermal demagnetization diagrams for gray dolomites from Douglas Dam before tilt correction and (d)-(f) corresponding plots (equal-area) of the migration of the vector endpoint during demagnetization [cf. Halls, 1976]. Conventions as in Figure 4. Open (solid) symbols in Figures 6d-6f represent projections on the upper (lower) hemisphere.

only limestones are suitable for chemical extraction techniques, as magnetite is typically more soluble in known solvents than dolomite or sphalerite. Representative limestone hand samples (about 1 kg each) were crushed, ground, and dissolved in buffered acetic acid of pH=5 [McCabe et al., 1983]. Magnetic extracts were obtained by pumping the insoluble residue slowly past a hand magnet. Although crude, this method yielded sufficient magnetic material from three samples. SEM of these extracts revealed the presence of abundant iron di-sulfides and varying amounts of sphalerite and iron oxide. The Fe-oxide particles ranged in size from 0.3 to 30 μm and occurred as almost perfect spheres (Figures 9a and 9b). Iron was the only cation detected by EDA (Figure 9c). Although the appearance of these spherules is similar to cosmic material, the lack of nickel, cobalt, or other cations seems to preclude a cosmic origin [Glass and Folger, 1978; McCabe et al., 1983, and references therein]. These observations are considered to be consistent with a diagenetic origin as proposed by McCabe et al. [1983].

Timing of Acquisition of the Characteristic Remanent Magnetization and the Age of Its Carriers

The present investigation did not reveal any directional differences in magnetization between the various lithologies sampled in different structural and geographic positions. Magnetizations in samples from both mineralized and unmineralized breccia bodies fail the conglomerate test convincingly. The main-stage sphalerite is older than the tectonism [Kendall, 1960], whereas the magnetization is thought to be contemporaneous with the Alleghenian folding, as argued below. Brecciation is thought to have occurred before or during sulfide mineralization [Haynes et al., 1987]. If these conclusions are all correct, then the magnetization is constrained to be younger than the ore, as the magnetization is postbrecciation.

The suggestion that the magnetization is younger than the ore is further corroborated by the magnetic characteristics of small samples taken in the laboratory from unoriented specimens

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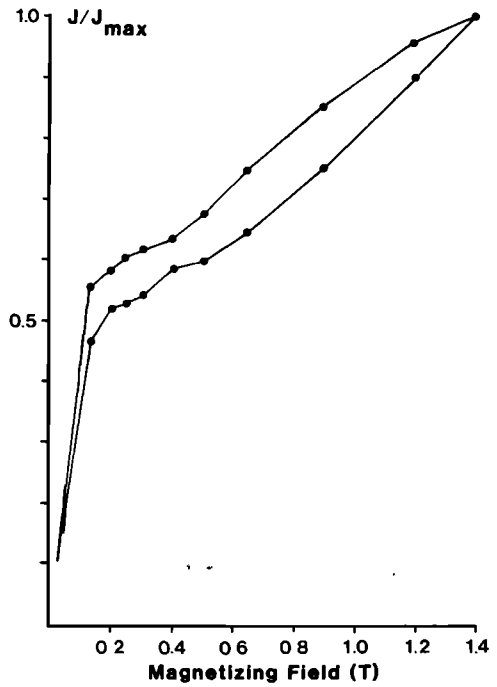


Fig. 7. Acquisition of isothermal remanent magnetization (IRM) of dolomite samples from the Douglas Dam locality.

of relatively pure sphalerite or secondary dolomite. Larger oriented samples of predominantly ore and secondary dolomite as well as these smaller unoriented mineral samples all have the same magnetic intensity per unit volume and similar coercivities and IRM acquisition curves. From this we conclude that the magnetization and its carriers are indiscriminately found in the host rock and in the ore and that therefore they postdate the mineralization.

In terms of the fold test, corrections for the tilt of the strata lead to a slight increase in the precision parameter k [Fisher, 1953]. However, the ratio of k_2/k_1 (precision parameter after unfolding divided by the precision parameter before unfolding) does not reach the critical value to be significant at the 95% confidence level [McElhinny, 1964]. Stepwise unfolding in 5% increments results in a maximum precision parameter k_m of 52.9 after about 60% of unfolding (Figure 10). Although the k_m/k_1 ratio of 1.67 is still not statistically significant at the 95% confidence level, the higher numerical maximum of k at 60% unfolding suggests a synfolding age for the magnetization. This argument is strengthened by a consideration of the directions before and after correction for tilt, given that the age of the folding is Late Carboniferous to Early Permian. If the magnetization were postfolding, then a mean direction without tilt correction (Table 1) would need to be considered as representative of a postfolding, i.e., post-Carboniferous, age. However, this direction (D/I = 158°/+32°) does not resemble the expected directions for post-Carboniferous times, and it is therefore very unlikely that the magnetization is younger than the folding. A prefolding age of the magnetization would require the full tilt correction to be applied. The resulting direction (D/I = 157°/+9°) does not, however, resemble the expected directions for Ordovician, Silurian, Devonian, or Early Carboniferous times [Miller and Kent, 1986; Van der Voo, 1987], which have steeper downward inclinations.

We conclude that the directions could possibly be Late Carboniferous and prefolding but that the best estimate of the age of the magnetization is Late Carboniferous and synfolding. A comparison of the paleopole, located at 40°N, 126°E as obtained from this best estimate (synfolding) direction, with the apparent polar wander path for cratonic North America (Figure 11) shows that a Late Carboniferous age for the magnetization fits very well.

The paleomagnetic results obtained from this study demonstrate convincingly that the observed magnetization is secondary and postdates brecciation. In addition, all our observations

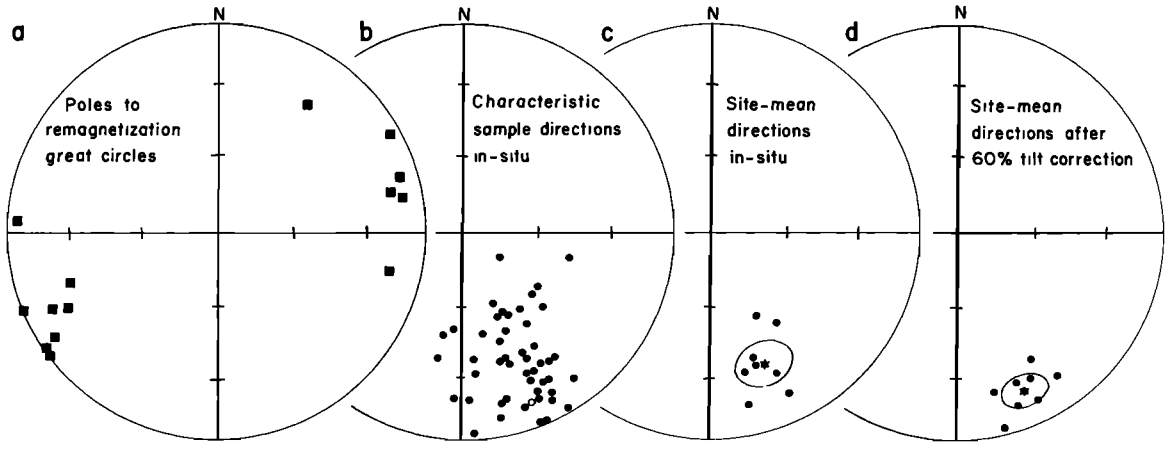


Fig. 8. Equal-area projections of (a) poles to remagnetization circles (squares) for Knox Group samples which show curved demagnetization trajectories [Halls, 1976], (b) characteristic sample directions obtained from principal component analysis [Kirschvink, 1980], and (c) and (d) site mean directions before and after 60% unfolding, respectively. Open (solid) symbols represent projections on the upper (lower) hemisphere. Resulting mean directions (stars) and the circle of confidence at the 95% probability level are also shown. See also Table 1.

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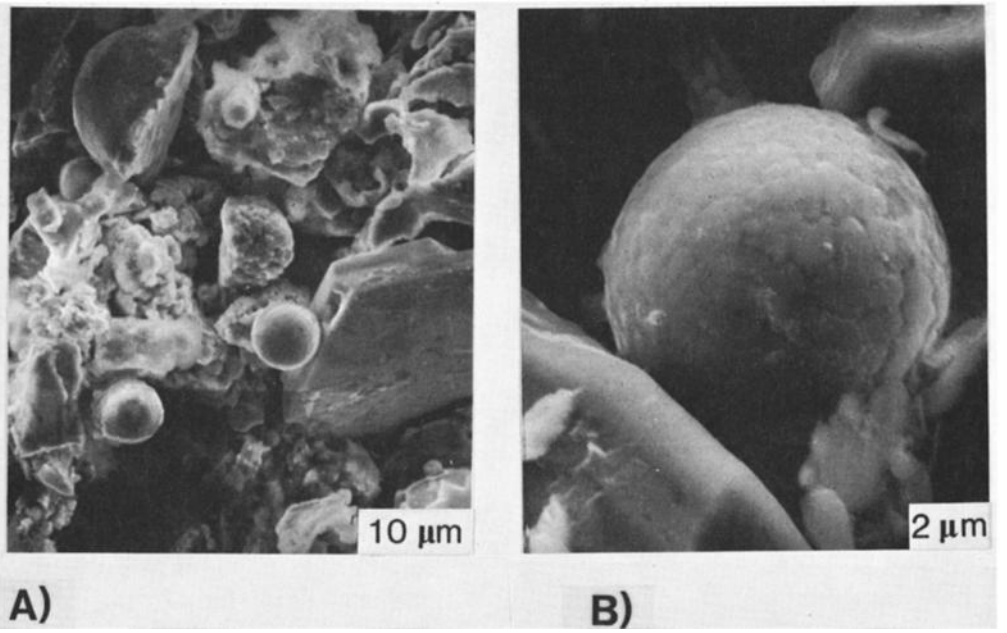


Fig. 9. (a) Scanning electron microscopy photograph of magnetite spheres extracted from limestone samples of the Ordovician Knox Group. Also shown is (b) an enlargement of one sphere and (c) its EDA spectrum, indicating iron as the only major element.

suggest that the magnetization postdates MVT ore formation in east Tennessee. The magnetization is inferred to be Late Carboniferous. But having determined a best estimate for the age of the magnetization as Late Carboniferous does not a priori imply that we have determined the age of the magnetite which carries this remanence. It is well known that magnetite in carbonate rocks can carry a viscous partial thermoremanent magnetization (VpTRM) much younger than the magnetites themselves, and Kent [1985] has suggested that such a VpTRM can have blocking temperatures and coercivities comparable to those of our study. Fluid inclusions in sphalerite and fluorite in east Tennessee document a maximum temperature of 170°C during mineralization

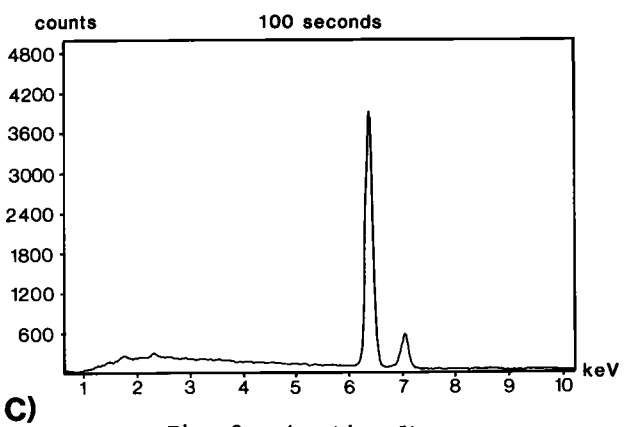


Fig. 9. (continued)

[Taylor et al., 1983; Zimmerman and Kesler, 1981]. The fact that these inclusions have not stretched or leaked implies that these temperatures were not exceeded since then. Whether these maximum temperatures are sufficient to activate a viscous remanence with laboratory blocking temperatures of up to 520°C can be determined from theoretical relationships [Walton, 1980; Moon and Merrill, 1986; Jackson and Van der Voo, 1986]. Given maximum unblocking temperatures for a magnetization isolated during the thermal demagnetization experiments, it is possible to estimate magnetic time-blocking temperature relationships, using thermal activation nomograms for single-domain magnetite [Middleton and Schmidt, 1982] based on Walton's [1980] theory for the acquisition of VpTRM. For our samples, a low-temperature present-day field component is removed by 300°C (Figure 4), and this maximum accurately matches the prediction from the Walton curves for a VpTRM acquired at room temperature during the last 700,000 years of the Brunhes Normal, thus lending credibility to this technique. The observed maximum laboratory unblocking temperature of 520°C for the Late Carboniferous magnetization requires heating at 230° for the duration of about 1 m.y. or 200°C for 100 m.y. Not only do these values exceed the estimate of Taylor et al. [1983], but they also would correspond to a Conodont alteration index (CAI) of 4 or higher, whereas CAIs of only 2.5-3 have been reported for the area under investigation [Epstein et al., 1977]. If this reasoning is correct, then the remagnetization observed in the Knox Group is very unlikely a VpTRM.

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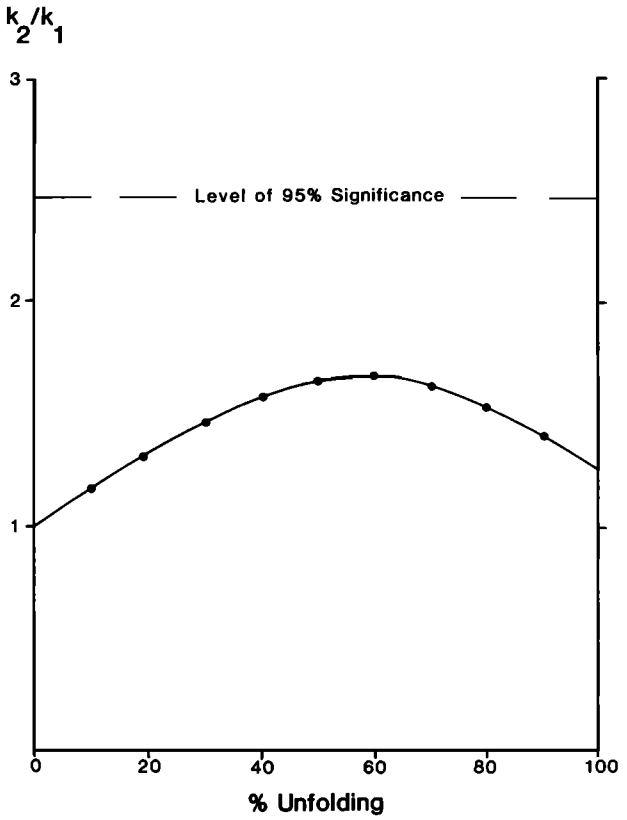


Fig. 10. Change of the k_2/k_1 ratio (precision factor after unfolding divided by the precision factor before unfolding) during stepwise unfolding.

The uniform distribution of magnetic intensities and behavior, regardless of rock type and the degree of mineralization (Figure 3), points to a pervasive formation of magnetite in the ore and all the carbonates. If the magnetite acquired its magnetization immediately, the latter would be a chemical remanent magnetization (CRM), possibly related to Alleghenian tectonic events. We argue that it is very likely that the magnetizations were acquired immediately, but even if that is not the case, magnetite formation must have been a secondary event. This secondary growth of magnetite is bracketed in time by the age of the mineralization and the age of the magnetization (Late Carboniferous).

Thus we can conclude that the magnetite in the Knox Group is of secondary origin and that the age of the magnetite precipitation is younger than that of the sphalerite mineralization. Ironically, then, the study has failed to constrain the age of mineralization in any significant manner but has instead succeeded in providing constraints on the age of the magnetites and the nature and age of their characteristic magnetization.

Iron to form the magnetite could have been introduced into the rock at the time of magnetite formation or it could have been derived from within the host rock. If the iron to form the magnetite was introduced into the Knox Group by a later solution, the magnetite should be distributed inhomogeneously in the rock and should concentrate in fractures and more permeable rock units. Our results, to date, indicate that the magnetite is distributed homogeneously, which suggests that the iron was derived in situ from the Knox Group host rocks. Churnet et al. [1982] have shown that the Knox carbonate rocks contain between 200 and 2000 ppm Fe in solid solution, with the highest concentrations in dolomite. In contrast, our magnetic data indicate that only about 1 ppm Fe is present in the rocks as magnetite. Thus recrystallization of very small amounts of Knox carbonate rock, probably along grain boundaries, would free more than enough iron to make the necessary magnetite.

Recrystallization of carbonate material to form the necessary magnetite should not cause a detectable change in the carbon and oxygen isotopic compositions of primary micritic carbonate, early calcite cements, and late sparry dolomite in these rocks. This is important because these carbonate components appear to retain original sedimentary and diagenetic isotopic compositions, which show no evidence of such a late recrystallization event (K.C. Lohmann, personal communication, 1987). The formation of magnetite, even in the sphalerite-bearing ore zones, suggests that the solutions that caused recrystallization were in contact largely with carbonate minerals. The formation of magnetite rather than other iron-bearing minerals apparently reflects the low activity of sulphur and high Ca/Fe ratio in the solutions, as well as a relatively low activity of oxygen, perhaps caused by reaction between the solutions and traces of organic material in the rocks.

Conclusions

A paleomagnetic mean direction (D/I = $157^{\circ}/18^{\circ}$) in Lower Ordovician rocks is observed to reside in magnetite of diagenetic origin. The magnetite and its magnetization postdate the formation of mineralized and unmineralized breccia bodies and are both of secondary origin. A fold test, although not significant at the 95% probability level, suggests a synfolding (Alleghenian) age for the acquisition of the magnetization. The resulting pole position (40°N, 126°E) is indistinguishable from the Late Carboniferous segment of the Paleozoic apparent polar wander path for North America and is significantly different from older Carboniferous or Devonian poles [Miller and Kent, 1986; Van der Voo, 1987]. Observed time-blocking temperature relationships and inferred paleotemperatures seem to preclude a widespread viscous-thermal origin for this remagnetization. Instead the post-ore character of the magnetite suggests a chemical remagnetization event related to the Alleghenian orogeny.

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Fig. 11. The pole position derived from this study (star) plotted with the North American apparent polar wander path [Van der Voo, 1987].

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